

TCC70xx MCU BSP

API Specification for Fault Management Unit

Rev. 0.10 [A]

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Preliminary version

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1 INTRODUCTION

This document describes the fault management unit device driver (hereinafter referred to as FMU driver) of TCC805x MCU BSP. For more detailed information on fault management, refer to Chapter 26 Fault Management Unit (FMU) in "*TCC70xx Full Specification*" [1].

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

FMU	Fault Management Unit
PMU	Power Management Unit
SMU	System Management Unit

3 SOURCE FILES

3.1 Location of Source Files

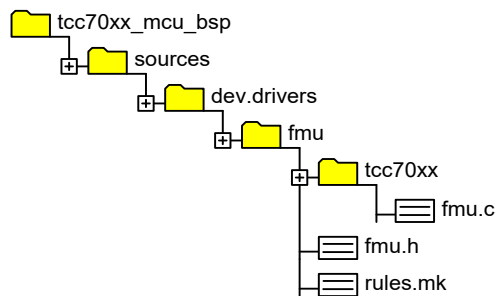


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- fmu.c
 - FMU driver source
- fmu.h
 - FMU driver header
- rules.mk
 - Rules used for build

4 ENUMERATIONS

4.1 FMUFaultid_t

Enumeration variables for fault source ID of FMU user interface API

Declaration

```
typedef enum FMUFaultid
{
    FMU_ID_ADC                = 0,
    FMU_ID_GPSB0              = 1,
    FMU_ID_GPSB1              = 2,
    FMU_ID_GPSB2              = 3,
    FMU_ID_GPSB3              = 4,
    FMU_ID_GPSB4              = 5,
    FMU_ID_GPSB5              = 6,
    FMU_ID_PDM0               = 7,
    FMU_ID_PDM1               = 8,
    FMU_ID_PDM2               = 9,
    FMU_ID_MC_CFG              = 10,
    FMU_ID_CR5_CFG             = 11,
    FMU_ID_CKC_CFG             = 12,
    FMU_ID_GPIO_CFG           = 13,
    FMU_ID_MID_CFG             = 14,
    FMU_ID_CMU_CFG             = 15,
    FMU_ID_SYS_SM_CFG          = 16,
    FMU_ID_X2X_MST0            = 17,
    FMU_ID_X2X_MST1            = 18,
    FMU_ID_X2X_SLV0            = 19,
    FMU_ID_X2X_SLV1            = 20,
    FMU_ID_X2X_SLV2            = 21,
    FMU_ID_SDM_VIDEO           = 22,
    FMU_ID_SDM_AUDIO           = 23,
    FMU_ID_SRVC                = 24,
    FMU_ID_PERI_CLK            = 25,
    FMU_ID_PLL1_CLK            = 26,
    FMU_ID_SRAM0_ECC           = 27,
    FMU_ID_SRAM1_ECC           = 28,
    FMU_ID_SFMC_ECC            = 29,
    FMU_ID_MBOX0_S_ECC         = 30,
    FMU_ID_MBOX0_NS_ECC        = 31,
    FMU_ID_MBOX1_S_ECC         = 32,
    FMU_ID_MBOX1_NS_ECC        = 33,
    FMU_ID_MBOX2_ECC           = 34,
    FMU_ID_OSC                 = 35,
    FMU_ID_VLD_CORE            = 36,
    FMU_ID_VLD_OTP             = 37,
    FMU_ID_WDT                 = 38,
    FMU_ID_PLL0_CLK            = 39,
    FMU_ID_CPU_CLK             = 40,
    FMU_ID_BUS_CLK             = 41,
    FMU_ID_CHIP_CFG_SOFT       = 42,
    FMU_ID_CHIP_CFG_CRC        = 43,
    FMU_ID_ROM_CRC             = 44,
    FMU_ID_FMU_CFG_SOFT        = 63
}
```

```
} FMUFaultId_t;
```

Description

Value	Description
<i>FMU_ID_ADC</i>	<i>ADC fault</i>
<i>FMU_ID_GPSB0</i>	<i>GPSB-0 fault</i>
<i>FMU_ID_GPSB1</i>	<i>GPSB-1 fault</i>
<i>FMU_ID_GPSB2</i>	<i>GPSB-2 fault</i>
<i>FMU_ID_GPSB3</i>	<i>GPSB-3 fault</i>
<i>FMU_ID_GPSB4</i>	<i>GPSB-4 fault</i>
<i>FMU_ID_GPSB5</i>	<i>GPSB-5 fault</i>
<i>FMU_ID_PDM0</i>	<i>PDM-0 fault</i>
<i>FMU_ID_PDM1</i>	<i>PDM-1 fault</i>
<i>FMU_ID_PDM2</i>	<i>PDM-2 fault</i>
<i>FMU_ID_MC_CFG</i>	<i>MCU configuration soft fault</i>
<i>FMU_ID_CR5_CFG</i>	<i>Cortex-R5 configuration soft fault</i>
<i>FMU_ID_CKC_CFG</i>	<i>Clock controller configuration soft fault</i>
<i>FMU_ID_GPIO_CFG</i>	<i>GPIO configuration soft fault</i>
<i>FMU_ID_MID_CFG</i>	<i>MID configuration soft fault</i>
<i>FMU_ID_CMU_CFG</i>	<i>Clock monitoring unit configuration soft fault</i>
<i>FMU_ID_SYS_SM_CFG</i>	<i>System safety mechanism configuration soft fault</i>
<i>FMU_ID_X2X_MST0</i>	<i>X2X master fault-0</i>
<i>FMU_ID_X2X_MST1</i>	<i>X2X master fault-1</i>
<i>FMU_ID_X2X_SLV0</i>	<i>X2X slave fault-0</i>
<i>FMU_ID_X2X_SLV1</i>	<i>X2X slave fault-1</i>
<i>FMU_ID_X2X_SLV2</i>	<i>X2X slave fault-2</i>
<i>FMU_ID_SDM_VIDEO</i>	<i>SDM fault-0</i>
<i>FMU_ID_SDM_AUDIO</i>	<i>SDM fault-1</i>
<i>FMU_ID_SRVC</i>	<i>SRVC fault</i>
<i>FMU_ID_PERI_CLK</i>	<i>Peripheral clock generator fault</i>

Value	Description
<i>FMU_ID_PLL1_CLK</i>	<i>PLL-1 output clock fault</i>
<i>FMU_ID_SRAM0_ECC</i>	<i>IMC RAM-0 ECC fault</i>
<i>FMU_ID_SRAM1_ECC</i>	<i>IMC RAM-1 ECC fault</i>
<i>FMU_ID_SFMC_ECC</i>	<i>SFMC RAM-1 ECC fault</i>
<i>FMU_ID_MBOX0_S_ECC</i>	<i>Secure MBOX-0 RAM ECC fault</i>
<i>FMU_ID_MBOX0_NS_ECC</i>	<i>Non-secure MBOX-0 RAM ECC fault</i>
<i>FMU_ID_MBOX1_S_ECC</i>	<i>Secure MBOX-1 RAM ECC fault</i>
<i>FMU_ID_MBOX1_NS_ECC</i>	<i>Non-secure MBOX-1 RAM ECC fault</i>
<i>FMU_ID_MBOX2_ECC</i>	<i>MBOX-2 RAM ECC fault</i>
<i>FMU_ID_OSC</i>	<i>XIN fault</i>
<i>FMU_ID_VLD_CORE</i>	<i>Core voltage fault</i>
<i>FMU_ID_VLD_OTP</i>	<i>OTPROM voltage fault</i>
<i>FMU_ID_WDT</i>	<i>Watchdog timer fault</i>
<i>FMU_ID_PLL0_CLK</i>	<i>PLL-0 output clock fault</i>
<i>FMU_ID_CPU_CLK</i>	<i>CPU clock fault</i>
<i>FMU_ID_BUS_CLK</i>	<i>Bus clock fault</i>
<i>FMU_ID_CHIP_CFG_SOFT</i>	<i>Chip configuration soft fault</i>
<i>FMU_ID_CHIP_CFG_CRC</i>	<i>Chip configuration CRC fault</i>
<i>FMU_ID_ROM_CRC</i>	<i>ROM CRC fault</i>
<i>FMU_ID_FMU_CFG_SOFT</i>	<i>FMU configuration soft fault (Only for IRQ_EN/CLR/STS)</i>

Remark

None

4.2 FMUSeverityLevelType_t

Enumeration variables for FMU severity level types

Declaration

```
typedef enum FMUSeverityLevelType
{
    FMU_SVL_LOW    = 0,
    FMU_SVL_MID    = 1,
    FMU_SVL_HIGH   = 2
} FMUSeverityLevelType_t;
```

Description

Value	Description
FMU_SVL_LOW	FMU generates an IRQ when a hardware fault occurs.
FMU_SVL_MID	FMU generates an FIQ when a hardware fault occurs.
FMU_SVL_HIGH	FMU generates an FIQ and External Error Indication Signal through GPIO-K[11] when a hardware fault occurs.

Remark

None

4.3 FMUErrTypes_t

Enumeration variables for FMU error types

Declaration

```
typedef enum FMUErrTypes
{
    FMU_OK                = 0,
    FMU_FAIL              = 1,
    FMU_INVALID_ID_ERR    = 2,
    FMU_RESERVE_ID_ERR    = 3,
    FMU_INVALID_SEVERITY_ERR = 4,
    FMU_ISR_HANDLER_ERR   = 5,
    FMU_DS_COMPARE_FAIL_ERR = 6,
    FMU_INTSET_ERR        = 7,
    FMU_INTEN_ERR         = 8,
    FMU_INTDIS_ERR        = 9
} FMUErrTypes_t;
```

Description

Value	Description
<i>FMU_OK</i>	<i>No error</i>
<i>FMU_FAIL</i>	<i>Unspecified FMU error</i>
<i>FMU_INVALID_ID_ERR</i>	<i>Invalid source fault ID value setting error</i>
<i>FMU_RESERVE_ID_ERR</i>	<i>Reserved source fault ID value setting error (45 to 62)</i>
<i>FMU_INVALID_SEVERITY_ERR</i>	<i>Invalid severity level value setting error</i>
<i>FMU_ISR_HANDLER_ERR</i>	<i>ISR Handler error</i>
<i>FMU_DS_COMPARE_FAIL_ERR</i>	<i>Double Storage comparison error</i>
<i>FMU_INTSET_ERR</i>	<i>FMU interrupt setting error</i>
<i>FMU_INTEN_ERR</i>	<i>FMU interrupt enable error</i>
<i>FMU_INTDIS_ERR</i>	<i>FMU interrupt disable error</i>

Remark

None

5 CALLBACK FUNCTIONS

5.1 FMUIntFnctPtr

When a hardware error corresponding to a specific fault source ID occurs, the registered callback function is called to notify the device driver.

Declaration

```
Typedef void (*FMUIntFnctPtr)
(
    void *    pArg
);
```

Parameters

Value	Description
pArg	[in/out] Callback function argument

Return

None

Remark

None

6 APIs

6.1 FMU_Init

Performs initialization as follows:

- Resets software of FMU.
- Sets the signal that notifies the hardware fault to the outside of TCC70xx.
- Registers IRQ/FIQ interrupt handler.

Declaration

```
FMUErrTypes_t FMU_Init
(
    void
);
```

Parameters

None

Return

Value	Description
FMUErrTypes_t	FMU Error type. Refer to FMUErrTypes_t .

Remark

None

6.2 FMU_IsrHandler

Configures the severity level (*uiSeverityLevel*) and callback function (*fnIntFnct*) with argument (*pArg*) for the given FMU ID (*uiId*).

Declaration

```
FMUErrTypes_t FMU_IsrHandler
(
    FMUFaultid_t      uiId,
    FMUSeverityLevelType_t uiSeverityLevel,
    FMUIntFnctPtr     fnIntFnct,
    void *            pArg
);
```

Parameters

Value	Description
uiId	[in] Fault source ID. Refer to FMUFaultid t .
uiSeverityLevel	[in] Severity Level. Refer to FMUSeverityLevelType t .
fnIntFnct	[in] ISR handler callback function. Refer to FMUIntFnctPtr .
pArg	[in] Argument

Return

Value	Description
FMUErrTypes_t	FMU Error type. Refer to FMUErrTypes t .

Remark

None

6.3 FMU_Set

Sets the severity level, EN register, and IRQ/FIQ/Fault mask register to detect the hardware fault based on the source ID received from device drivers.

Declaration

```
FMUErrTypes_t FMU_Set
(
    FMUFaultid_t    uiId
);
```

Parameters

Value	Description
uiId	[in] Fault source ID. Refer to FMUFaultid t .

Return

Value	Description
FMUErrTypes_t	FMU Error type. Refer to FMUErrTypes t .

Remark

None

6.4 FMU_IsrClr

Clears the interrupt status of FMU driver to stop generating the same fault interrupt.

Declaration

```
FMUErrTypes_t FMU_IsrClr
(
    FMUFaultid_t    uiId
);
```

Parameters

Value	Description
uiId	[in] Fault source ID. Refer to FMUFaultid t .

Return

Value	Description
FMUErrTypes_t	FMU Error type. Refer to FMUErrTypes t .

Remark

None

6.5 FMU_Deinit

Releases the IRQ/FIQ interrupt handler registered in GIC400.

Declaration

```
FMUErrTypes_t FMU_Deinit
(
    void
);
```

Parameters

None

Return

Value	Description
FMUErrTypes_t	FMU Error type. Refer to FMUErrTypes t .

Remark

None

7 HOW TO USE FMU DRIVER

7.1 Initialize FMU Driver

Initialize the FMU driver by executing *FMU_Init()*.

```
int BSP_Init ()
{
    I2C_Init();
    FMU_Init();           //FMU driver initialization
    GPSB_Init();
    ADC_Init(ADC_MODE_NORMAL);
}
```

7.2 Example: Using FMU in Other Device Drivers

The following example shows how to configure the FMU by using APIs with *FMU_ID_GPSB0*.

```
static void GPSB_FmuIsr0(void * pArg)
{
    FMUErrTypes_t ret = FMU_FAIL;

    ret = FMU_IsrClr(FMU_ID_GPSB0);           //interrupt clear

    return ret;
}

static int32 GPSB_EnableFmu (void)
{
    FMUErrTypes_t ret = FMU_FAIL;

    ret = FMU_IsrHandler(FMU_ID_GPSB0, FMU_SVL_LOW, GPSB_FmuIsr0, NULL); //register callback func
    ret = FMU_Set(FMU_ID_GPSB0);           //FMU set for HW fault

    return ret;
}
```


8 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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